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The integral in this exercise converges. Evaluate the integral without using a table.

$$\int_0^{324} \frac{dx}{\sqrt{324-x}}$$

Recall that, for Type II improper integrals, if $f(x)$ is continuous on $[a,b)$ and discontinuous at b , then $\int_a^b f(x)dx = \lim_{c \rightarrow b^-} \int_a^c f(x)dx$. So

$$\int_0^{324} \frac{dx}{\sqrt{324-x}} = \lim_{c \rightarrow 324^-} \int_0^c \frac{dx}{\sqrt{324-x}}.$$

Integrate.

$$\lim_{c \rightarrow 324^-} \int_0^c \frac{dx}{\sqrt{324-x}} = \lim_{c \rightarrow 324^-} \left[-2\sqrt{324-x} \right]_0^c$$

Substitute the limits of integration.

$$\lim_{c \rightarrow 324^-} \left[-2\sqrt{324-x} \right]_0^c = \lim_{c \rightarrow 324^-} \left(-2\sqrt{324-(c)} \right) - \left(-2\sqrt{324-(0)} \right)$$

Evaluate the limit of the expression to find the value of the integral.

$$\begin{aligned} \lim_{c \rightarrow 324^-} \left[-2\sqrt{324-c} \right] - \left[-2\sqrt{324-0} \right] &= 0 - (-36) \\ &= 36 \end{aligned}$$

Therefore, $\int_0^{324} \frac{dx}{\sqrt{324-x}} = 36.$